

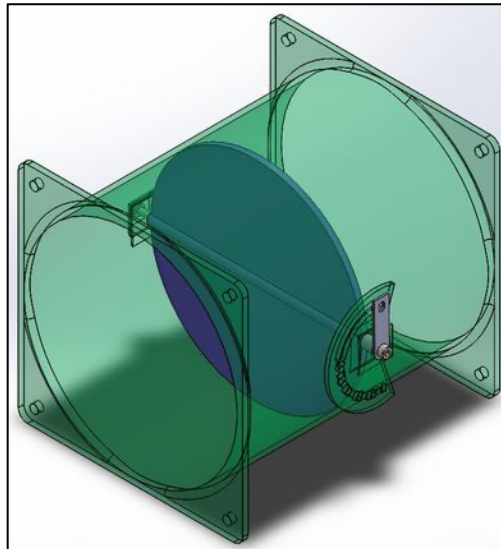


UNIVERSITY
OF WOLLONGONG
AUSTRALIA

LAB REPORT

Subject: MECH470/970

Motor Modelling



Week 10

Student Name	Student Number
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1. Overview

This lab involves developing a Simscape model of a DC servo motor that actuates an oxygen valve in a miniature rocket engine. The benefit of Simulink's Simscape Toolbox for intuitive modelling without having to derive mathematical functions is highlighted. The model is progressively enhanced in each task by parameterising the motor and observing its open-loop response. Finally, a variable torque load is applied to simulate the aerodynamic resistance on the butterfly valve to understand how real-world loads influence performance. Comparisons between the modelling stages are presented and discussed below.

2. Results & Discussion

Developing the motor model

The modelled motor drive based on *Appendix A*, *Figure 2* is displayed in *Figure 1*.

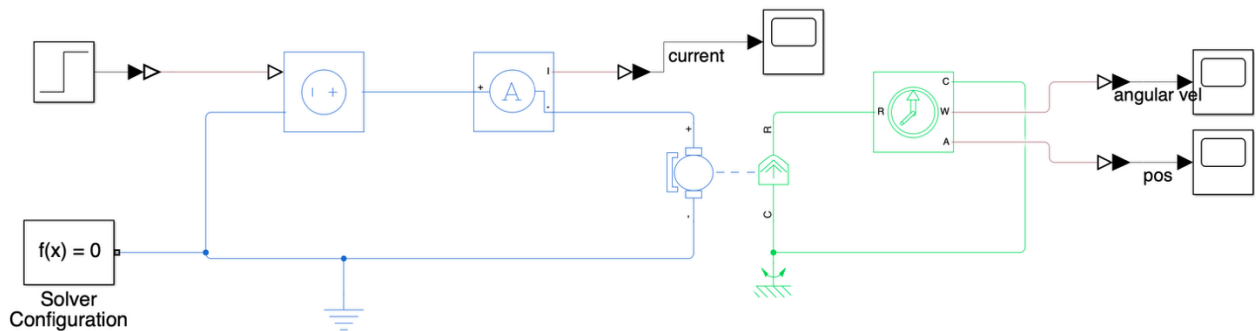


Figure 1 - Electro-mechanical model of motor system

A plot of the motor's open-loop position, speed, and current drawn to a step input for 10 seconds is displayed in *Appendix A*, *Figure 4*. The position keeps increasing infinitely at a velocity of about **1454 rad/s** as there is no resistive force like friction, by the end of 10 seconds, it has reached a value of **13080 radians**. The velocity remains unchanged upon reaching the steady value (at 1.06 s) due to the same reason. The plot of the current shows a sudden increase to **0.256 A** upon applying a step input, then a sudden decrease to 0 as the current is drawn instantly by the motor due to its lack of resistance.

Motor Parameterisation and Open Loop Responses

The following specifications are added to the MT-2240-AMYAN servo motor drive:

- Armature Resistance: 4Ω
- Armature Inductance: 7.7 mH
- Torque Constant: 0.115 Nm/amp
- Rotor Inertia: 0.05 kgm^2
- Rotor Damping: 0.012 Nm/(rad/s)

A plot of the motor's open-loop position, speed, and current drawn with the motor parameters to a step input for 10 seconds is displayed in *Appendix A*, *Figure 5*. With the added resistance, damping and other parameters, the position and velocity increase much slowly with a low step input voltage as only a portion ($\sim 78.5\%$) of the current is drawn into the motor (instead of all the current). However, the higher the step input, the more current the motor would have to exponentially increase the position and velocity of the motor.

Variable Load Function

The rotational torque exerted by the fuel flow against the butterfly valve is mathematically modelled adding the $T = -0.3 |\sin(p)|$ as an ideal torque source as displayed in *Figure 2*.

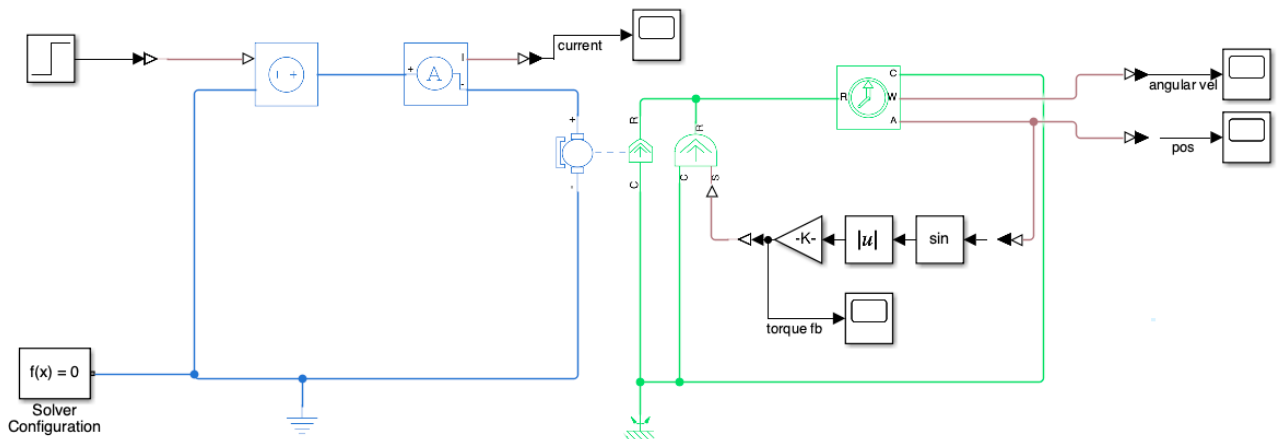


Figure 2 - Electro-mechanical model of motor system with variable load function

A plot of the motor's open-loop position, speed, and torque feedback with the motor parameters to a step input for 10 seconds is displayed in *Appendix A, Figure 6*. The velocity over time now varies because of the torque exerted by the fuel, as a result, the position increases but not at a consistent rate. The torque feedback plot displays how the force exerted (and therefore torque experienced) varies based on the position of the butterfly valve.

Full system model

The operation of the system is visually observed by connecting the valve assembly. The system operates as expected with the valve rotating at different rates based on its position and therefore torque experienced.

3. Conclusion & Improvements

Feedback and Suggestions

The system provides a good analysis on how the different parameters affect the response, but it doesn't quantify how strongly each parameter, such as the torque constant, rotor damping or inertia, influences the motor response. A sensitivity study and testing with different values for parameters is required to gain a better understanding.

The open-loop model lacks mechanical losses, leading to unrealistic infinite acceleration under a constant voltage; adding friction blocks would produce more physically accurate behaviour. However, the Simscape model successfully demonstrates the electromechanical behaviour of a DC servo motor both in unloaded and torque-loaded conditions.

4. Appendix A

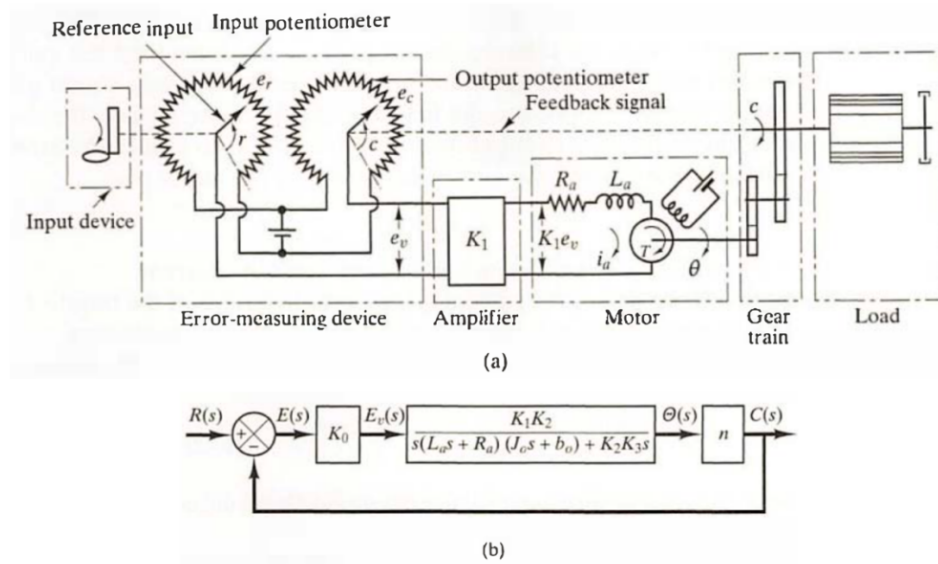


Figure 3 - (a) Position-controlled servo motor schematic and (b) derived transfer function for the servo motor.

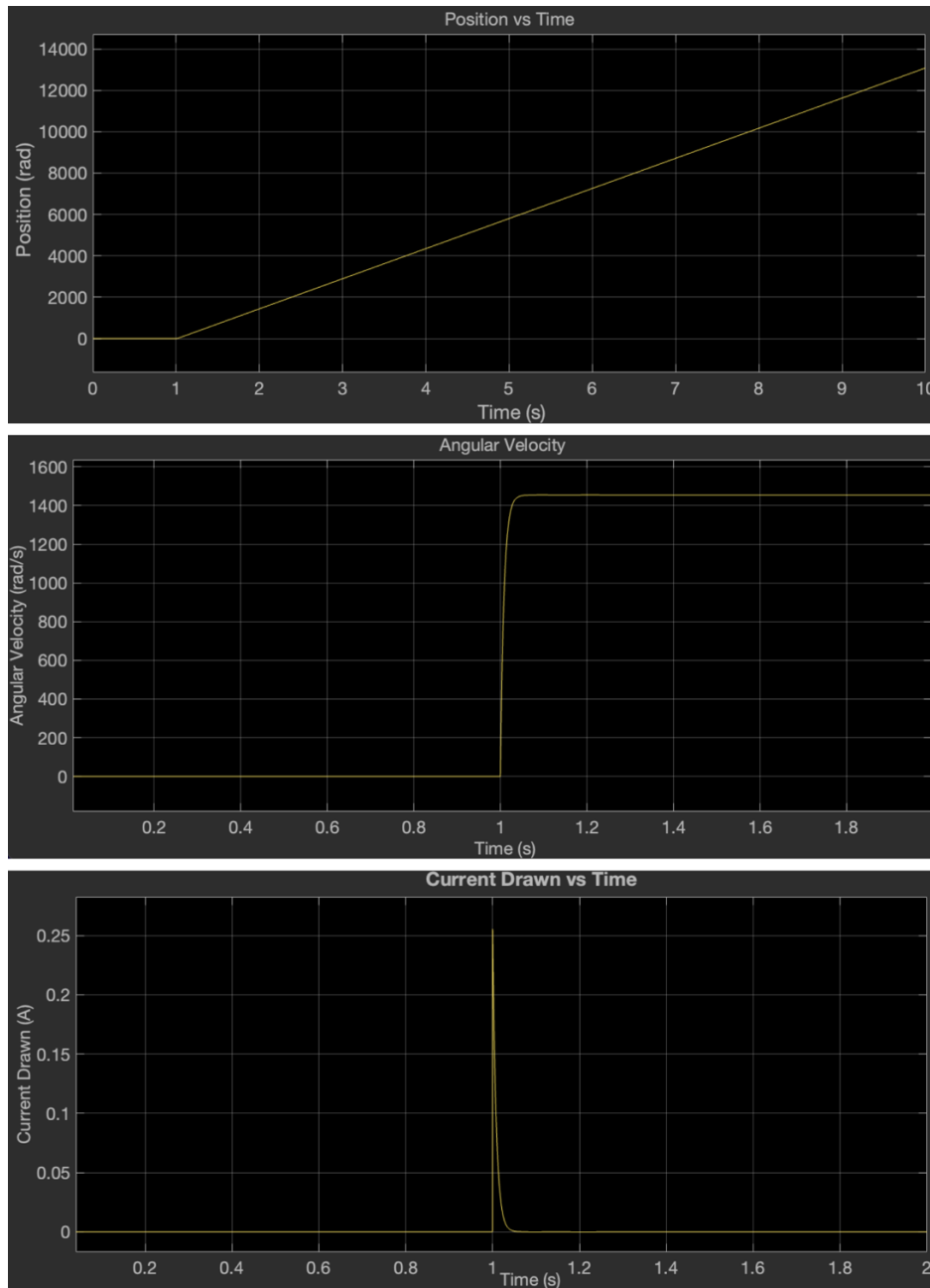


Figure 4 – Motor's open-loop position (top), speed (middle), and current drawn (bottom)

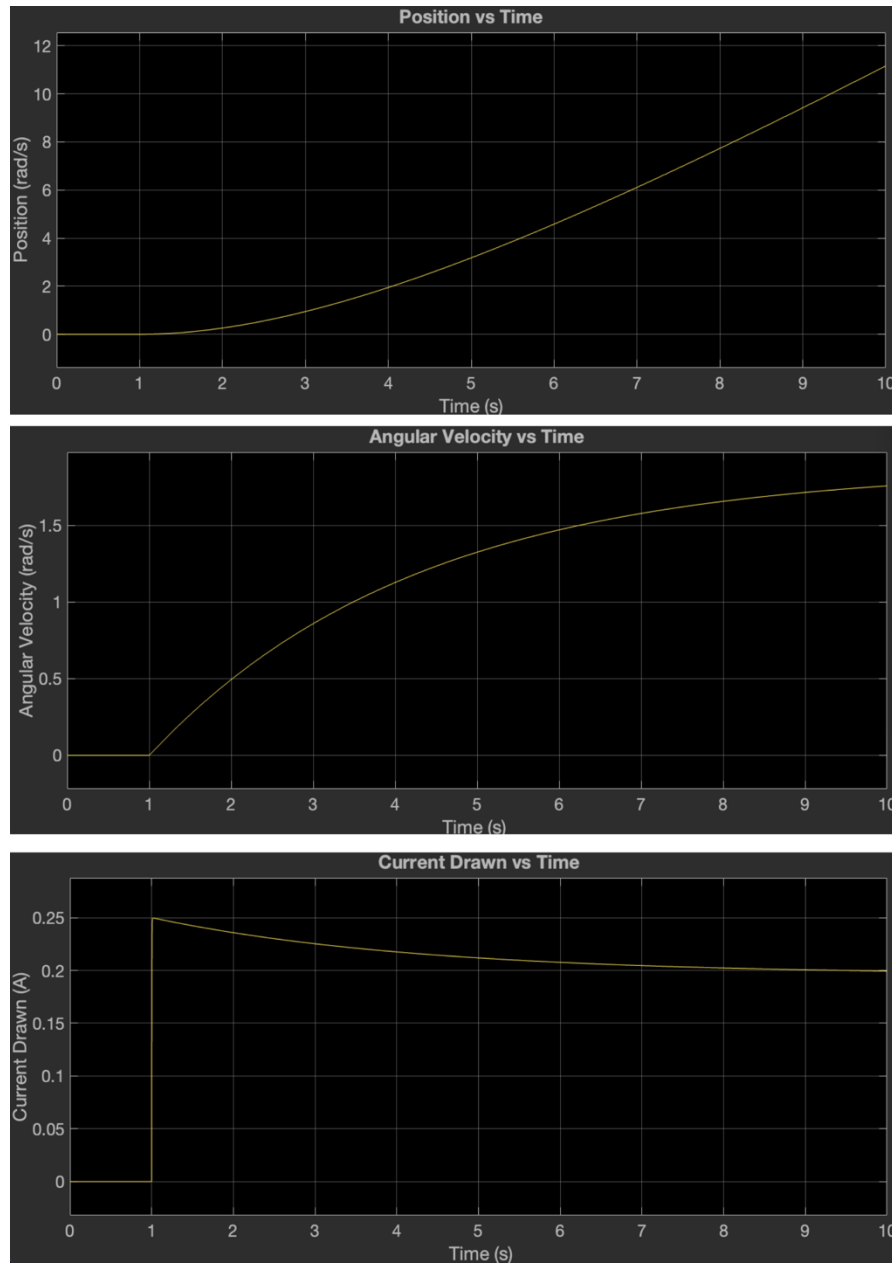


Figure 5 - Motor's open-loop position (top), speed (middle), and current drawn (bottom) after motor parameters

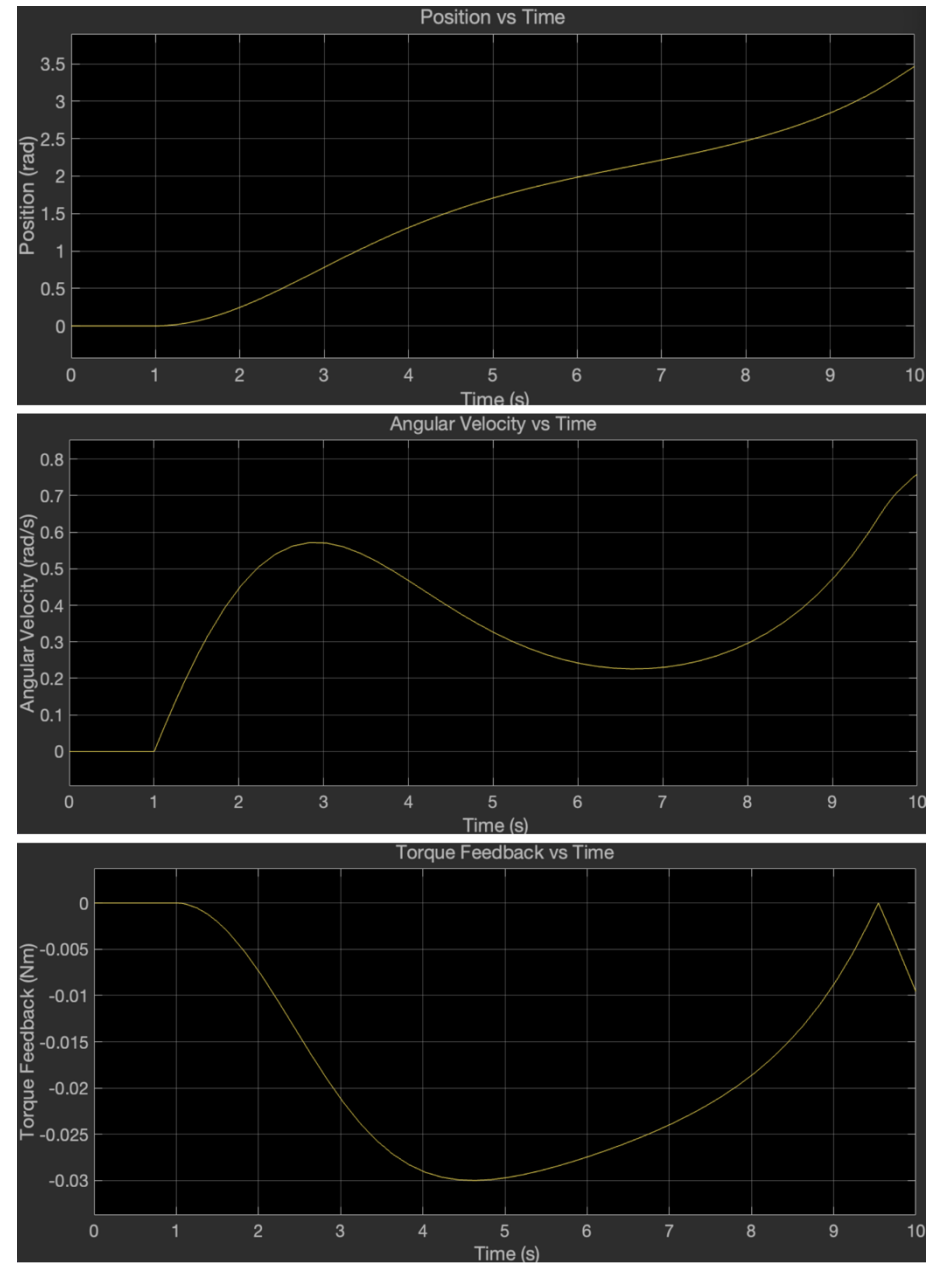


Figure 6 - Motor's open-loop position (top), speed (middle), and torque feedback (bottom) after variable load function